3-14 Sepsis v.1

Severe sepsis (hypotension persisting after initial fluid challenge of 30ml.kg^{-1} or blood lactate concentration ≥ 4mmol.l^{-1} if infection most likely underlying cause) or septic shock (sepsis with end organ dysfunction).

START

1. Call for help and inform theatre team of problem.
2. Increase FiO_2, consider reducing anaesthetic agent and intubate patient.
3. Give crystalloid i.v.:
   - Adult: at least 30 ml.kg^{-1} (Box A, Box B).
   - Child: at least 20 ml.kg^{-1} (Box C).
4. Take bloods including blood gas, lactate, FBC, U&Es, coagulation and cultures.
5. Give empiric intravenous antimicrobials within 1 h (seek microbiology advice).
6. Consider whether indwelling devices could have caused a septic shower.
7. If patient is not improving proceed to the next steps.
8. Insert central and arterial access lines. Check serial lactates.
9. Start noradrenaline to achieve mean arterial pressure ≥ 65 mmHg (Box D).
10. Insert urinary catheter and record hourly urine output.
11. Consider monitoring cardiac output to further aid fluid and vasopressor therapy.
12. Identify source of sepsis, consider source control and send source cultures if possible (eg. surgical site, urine, broncho-alveolar lavage).
13. Discuss whether appropriate to abandon or limit surgery.
14. Discuss ongoing management plan with intensive care team.

Box A: FLUID THERAPY

- Crystalloids initial fluid of choice in severe sepsis and septic shock.
- Greater than 30 ml.kg^{-1} of crystalloid may be required in some patients.
- Continue fluid challenge if haemodynamic improvement.
- Hydroxyethyl starches should not be used.

Box B: SET PHYSIOLOGICAL GOALS

- Central venous pressure.
- Mean arterial pressure.
- Urine output.
- Central venous (superior vena cava) or mixed venous saturation.

Box C: PAEDIATRIC CONSIDERATIONS

- Goals: capillary refill time (CRT) ≤ 2 secs, normal BP for age, normal peripheral pulses, warm extremities, urine >1 ml.kg^{-1}.hr^{-1}, $SCVO_2$ >70%.
- Give 20 ml.kg^{-1} initially up to or over 60 ml.kg^{-1} fluid until goals or unless rales or hepatomegaly develops.
- Begin peripheral inotropic support pending central/intraosseous access.
- If warm shock (↑HR, ↓BP) start noradrenaline.
- If cold shock (↑HR, ↓CRT) start dopamine and, if resistant,adrenaline.

Box D: DRUG THERAPY

- **Noradrenaline (NA)** as first choice vasopressor.
- **Adrenaline** added to noradrenaline when additional agent needed.
- **Vasopressin** 0.03 units.min^{-1} added to ↑MAP or ↓noradrenaline need.
- **Dobutamine** up to 20 µg.kg^{-1}.min^{-1} if evidence of myocardial dysfunction or ongoing signs of hypoperfusion despite adequate MAP and adequate intravascular volume.
- **Hydrocortisone** if unable to restore haemodynamic stability.

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